

Name: _____

ANGLES AND GEOMETRY, (ANGLES IN POLYGONS)

Date: _____

LO: I can find interior and exterior angles of regular and irregular polygons.

Polygon angle rules

The sum of **interior angles** of a polygon with n sides is **$(n - 2) \times 180^\circ$** .

Exterior angles: The exterior angles of any convex polygon always sum to 360° . Each exterior angle of a regular polygon = $360^\circ \div n$.

Key facts

●	Triangle: $(3-2) \times 180 = 180^\circ$	angle sum
●	Quadrilateral: $(4-2) \times 180 = 360^\circ$	angle sum
●	Pentagon: $(5-2) \times 180 = 540^\circ$	angle sum
●	Hexagon interior sum = 600°	should be 720°
●	Exterior angles sum to 180°	always 360°

Key idea

Interior angle sum = $(n - 2) \times 180^\circ$. Exterior angles always sum to 360° .

$$\text{Sum} = (n - 2) \times 180^\circ$$

$n = \text{number of sides}$

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - interior angle sum

Find the sum of interior angles of a hexagon.

$$\begin{aligned} \text{A hexagon has 6 sides, so } n &= 6 \\ (6 - 2) \times 180 &= 4 \times 180 \\ &= 720^\circ \end{aligned}$$

$$= 720^\circ$$

Always subtract 2 from the number of sides first.

Example 2 - one interior angle (regular)

Find one interior angle of a regular octagon.

$$\begin{aligned} \text{Sum} &= (8 - 2) \times 180 = 1080^\circ \\ \text{Regular: all angles equal} \\ 1080 \div 8 &= 135^\circ \end{aligned}$$

$$= 135^\circ$$

For regular polygons, divide the total by the number of sides.

Example 3 - exterior angle

Find one exterior angle of a regular pentagon.

$$\begin{aligned} \text{Exterior angles sum to } 360^\circ \\ \text{Regular: all exterior angles equal} \\ 360 \div 5 &= 72^\circ \end{aligned}$$

$$= 72^\circ$$

Exterior angle + interior angle = 180° (check: $108 + 72 = 180$).

Example 4 - finding missing angle

A pentagon has angles 100° , 120° , 130° , 90° and x . Find x .

$$\begin{aligned} \text{Sum} &= (5 - 2) \times 180 = 540^\circ \\ 100 + 120 + 130 + 90 &= 440^\circ \\ x &= 540 - 440 = 100^\circ \end{aligned}$$

$$x = 100^\circ$$

Subtract the known angles from the total angle sum.

YOUR TURN – PRACTICE (deliberate practice)

1. Interior angle sums

Find the sum of interior angles.

- a) Triangle (3 sides)
- b) Pentagon (5 sides)
- c) Heptagon (7 sides)
- d) Decagon (10 sides)

2. One angle of a regular polygon

Find one interior angle of each regular polygon.

- a) Equilateral triangle
- b) Square
- c) Regular hexagon
- d) Regular nonagon (9)

3. Exterior angles

Find one exterior angle of each regular polygon.

- a) Regular triangle
- b) Regular quadrilateral
- c) Regular hexagon
- d) Regular decagon

4. Missing angles

Find the missing angle x .

- a) Triangle: 60° , 80° , x
- b) Quadrilateral: 90° , 85° , 110° , x
- c) Pentagon: 100° , 110° , 120° , 130° , x
- d) Hexagon: 130° , 140° , 100° , 110° , 120° , x

5. Mixed polygon problems

Answer each question.

- a) How many sides if interior sum = 1260° ? b) Each exterior angle is 40° . How many sides? c) Interior angle of regular polygon = 156° . How many sides? d) Sum of exterior angles of a 100-sided polygon? e) Interior angle of regular 12-sided polygon?

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) Interior angles of a hexagon sum to 720° ☐

Correct answer: _____

Reason: _____

b) Exterior angles always sum to 180° ☐

Correct answer: _____

Reason: _____

c) A regular polygon has all angles equal ☐

Correct answer: _____

Reason: _____

d) Interior angle sum of a quadrilateral is 180° ☐

Correct answer: _____

Reason: _____

e) Exterior angle = $360^\circ \div n$ for regular polygons ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) The interior angle of a regular polygon is **5 times** its exterior angle. How many sides does the polygon have? Show your working.

Expression: _____

Simplified: _____

b) Three angles of a quadrilateral are in the ratio **2 : 3 : 4**. The fourth angle is **120°** . Find each of the other three angles.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

